A hard copy of this is in Bowie8[faber].

June 8, 1990

Dear Jim:

I will talk to you on the phone, but here is a written core-dump and checklist of the items that I was currently pursuing and which you need to follow up. I have been keeping Jon Holtzman in the loop, so he can fill you in on details, and also describe the materials that are included in the file folders under this memo.

1) (Almost) daily routine:

08:30 AM Daily schedule meeting in STOCC conference room near Mission Control Room. Overall discussion of upcoming scheduled events. Seems to be mainly a reporting session rather than a strategy session.

09:00 AM SMS planning session. Very important! This is where the planning and strategy really get discussed. Attendees include John Campbell (SMS director), Gene Oliver (overall project leader here, it seems), Domenic Tenerelli, Charlie Jones, and there is usually a conference call with Roger Doxey at the end of the meeting. WFPC pictures are discussed here, and need to be planned/defended. The time and place of this meeting vary at the last minute, so you need to ask around early in the AM to find out. Location on most days is said to be in the same room as the previous meeting, but I have found it most often in another room at the end of the narrow corridor near the MOC that has all the IDT mailboxes in it. It takes place roughly every other day, when the next SMS is up for consideration. Its occurrence is also announced on the 11:00 meeting schedule (note, all times there are in GMT!).

John Campbell said to me today that he expected that the WFPC would be an integral player in focussing the OTA and that he (now) requested us to attend these SMS planning sessions on a regular basis.

A crucial SMS planning meeting was held on Sat. AM, which only Jon could attend. Ask him what happened.

11:00 AM Regular 11:00 meeting held on the second floor of Bldg. 3 run by Joe Ryan in which all segments of the Project report. Each time we take a picture, I bring some visual material and give a 2-minute summary of the results. Ryan has come to expect this without actually saying so. The WFPC results cheer everybody up, so it's important to bring something and say something.

2) Upcoming meeting: Performance Monitoring and Assessment Weekly Meeting #6, Tues. Jun. 12, Bldg. 8, Rm 303, run by Kalinowski. Minutes of the last meetings are attached, to give you some idea of the written report that you are expected to bring to this. I

missed the meeting last week (we were gently scolded) because I had never heard of it and didn't realize I was supposed to go. Bring some pictures, too. Suggested outline of our report to date on activites over the last two weeks is at the bottom of this memo.

3) Current status of WFPC picture taking:

There are currently two kinds of SMS's, which have been alternating. One is a Bootstrap SMS, which contains some portion of the procedure needed to focus the OTA (e.g., Bootstrap B, Phase 3, which we are currently in for a second try), and Science Instrument SMS's, which are used to put the instruments through their OV paces and also as filler and thinking time between the Bootstrap SMS's. At present:

a) There is no standing mandate to take any WFPC pictures during SI SMS's.

b) There is a standing mandate to take the Lauer sequence (10s, 100s, F547W) before and after any mirror adjustments in Bootstrap SMS's. These should all be taken with the 20" slew (target position 139.29470, -59.09393), not with the position that puts iota Car in wavefront sensor 3 (139.30556,-59.09329). The slew puts a fairly bright (15.0) mag star in the center of Chip 5, which will tell us if the kidney is indeed a filled ring (see below). It was the failure to include this slew that spoiled our last picture (Friday AM).

c) There is no mandate as yet to use the Burrows-Westphal sequence (see attached exposure logsheet) at any time whatsoever. This sequence has more F547M exposures than the Lauer sequence (of various lengths), plus a UV and an R exposure. There are also some FGS measurements, which I did not review and can't justify. There are two targets: a primary target with a 12.2 mag bright star that will give UV info, and a secondary target that has a lot of nice 15th mag stars that will give a good map of the PSF over the field, but no UV info. The primary target involves a slew from the default position (iota Car in Wavefront sensor 3) of ~20', while the secondary target is a slew of only 55" away. So far, I don't know whether this sequence (using either target) is included in the next SMS's (Bootstraps phase 4 and 5). This will be decided in the Sat AM SMS meeting, which Jon is attending. Check with him..

WE ARE CURRENTLY NOT GETTING ENOUGH INFO FROM THE PICTURES BEING TAKEN IN THE DEFAULT FIELD (NO SLEW). THE SYMMETRY OF THE KIDNEYS IS A KEY POINT. I ATTACH 9X9 MOSAICS OF THE IMAGES MADE BY ROGER LYNDS. NOTE THE APPROXIMATE CIRCULAR SYMMETRY ABOUT THE CENTER OF EACH OF THE CHIPS. SEE JON H FOR MORE DETAILS ON THESE MOSAICS. WE CLEARLY NEED TO FILL OUT THE INFORMATION IN THE POORLY SAMPLED LOCATIONS BY HAVINF A MUCH RICHER, WELL DISTRIBUTED STAR FIELD. THE BEST CANDIDATE FOR SUCH A FIELD IS THE 55" SLEW POSITION IN THE BURROWS-WESTPHAL PROPOSAL, BUT THE BRIGHT-STAR PRIMARY TARGET IN THIS PROPOSAL SHOULD ALSO BE PRETTY GOOD. NO ONE IN THE SYSTEM EXCEPT PERHAPS LYON REALLY UNDERSTANDS THE UTILITY OF THESE IMAGES, SO THERE ARE NO POWERFUL ADVOCATES. C. JONES COMES CLOSEST. DOMENIC TENERELLI HAS ALSO BEEN SUPPORTIVE OF MORE WFPC DATA AND HAS SECRETLY BEEN EGGING ME ON.

3) Major issues: I will explain these in more detail over the phone. This is just a list.

a) How best to interact with the system to lay the groundwork in the next few days to plan an alternative plan for focussing the OTA using the WFPC (and maybe FOC). Gene Oliver said yesterday he would give the Hughes folks exactly

one chance to execute Bootstrap B Part 4 (upcoming SMS early next week) and then he would start to find some other way to move the secondary mirror. The Hughes folks are being stymied by a combination of terminator jitter, plus the inability of the scheduling system to place their wavefront observations in the small window of the orbit free from jitter. I really feel sorry for them. Bob Basedow is a smart guy, but he's fighting with both hands behind his back. Gene Oliver is sounding increasingly desperate in the last two days. The WFPC is the next logical choice for help, so we'd better start getting ready.

Various suggestions to do this:

Convene a Tiger Team: (Westphal, Vaughn, Lyon, Burrows, Schroeder, Gunn, Trauger, Roddier(?), Basedow, Fastie(?), myself?, others??). You will doubtless have additional nominations, but in the event that more people are desirable, I would suggest Joe Miller of Lick, a real hands-on instrument builder/optician/astronomer and Director of the Lick shops, and Jerry Nelson and Terry Mast, who fathered the segmented Keck telescope and who are probably the world's experts on mirror actuators (should that knowledge become necessary). (Intellectually, Jerry N. is another Jim Gunn. If I had a really tough problem in physical optics, I'd turn to him. Also a real can-do guy. Problem: he and Terry are probably both up to their necks in commissioning the Keck telescope right now.)

Prepare a proposal involving WFPC pictures coupled with actuator motions of the secondary mirror and have it run through the STScI scheduling preparation software so that it is easy to implement. Would also need technical help from the system on TDRSS, the SAA, and the actuators. Burrows could help prepare and process the proposal at the STScI, as he did the Burrows-Westphal proposal.

Have a talk with Gene Oliver (recommended by Kalinowski).

b) How to provide "astronomical review" of the SMS's so that the WFPC exposures are properly implemented. We lost an exposure last night (Friday AM) through bad communication (a required slew was left out of the SMS by mistake). I have talked to Dave Gallagher about possibly using him to do more than review just the health and safety of WFPC. This is a really critical problem if we start using the camera in conjunction with highly technical observatory maueuvers, which in a way, has already started. John Campbell as much as said he expected us to do this now. I am very worried that we are going to start being held responsible for the accuracy of the SMS's without having the requisite knowledge of the whole Observatory and without (so far) being given the necessary technical education and support by the system. Accumulating the relevant knowledge is still a battle for me -- maybe with your entree into the system it will be easier.

c) Continue to coordinate optical modelling by Vaughn, Burrows, Trauger (he has an important action assignment), and Lyon. At the moment, all these people are communicating mainly through me. This should not continue!

d) Continue to interact with Burrows in developing and/or screening any new proposals to take images. For better or worse, we are in bed with him. He needs technical advice and review on how to take WFPC images. Griffiths is helping. Burrows can run proposals

through the STScI screening system, which makes them much more acceptable to Doxey.

4) Medium issues:

Get the system to appreciate the need for a simple focus run. Lyon (the PE optical modeller) prefers to tilt the secondary. thinks he understands what's wrong. Everyone else (Trauger, Faber, Vaughn, Lynds) recommends pictures at different despace. Lyon's model is a tilted OTA secondary that is sending light through the PC at a slanted angle. I disagree -- model doesn't give images with right symmetry (see 9x9 mosaics by Roger Lynds).

Look ahead and understand the FOC optics. It may also be used for focussing.

Look ahead and understand the PM actuators and what they can do. My nightmare worry is severe spherical aberration. How well can they correct this?

5) One-shot items that are pending:

Files of backup materials for the N188 and iota Car images are under this memo. Especially valuable is the polaroid print of the field we are using near iota Car. Overlay of PC goes with it, oriented at approximately 45 deg to north. 55" slew field is to immediate SW of the default field (ask Jon to show you where this is). I don't have a photo of the primary target field. Should get one.

The local sky survey is the SRC J survey, which is in the office of Klinglesmith-Nieder (locked up) on the ground-floor of Bldg. 21. If after hours or on a weekend, would need to get a security guard to let you in. Should probably talk to Nieder (K. is on travel) to arrange after-hours entry. The northern PSS is on paper prints in a file cabinet near the IUE area (same floor, down the hall). The Polaroid copy camera is right there. We have two boxes of film here -- take with you, as they are running low. There is someone at the IUE almost all the time (24 hrs/day).

Burrows-Westphal exposure sequence is under this memo.

Blueprints of the WFPC are coming in the mail - one set for us, one set for Chris Burrows. Arrange xfer to him immediately when they arrive.

I have asked Seidelman to check out the astrometry possibilities in relevant fields: the iota Car default field, the 55" slew target, and the 20' slew target. Some faxes are included under this memo. He stands ready to compute positions if we need them.

We still don't have a k-spot exposure for the PC. Tod asked Shawn to check the proper exposure by looking at the TV6 data. The last set was overexposed, but that may have just been due to the wrong orientation of the pyramid.

Review OV activities to see what's still undone.
I have drafted a memo (attached) for revisions to our OV proposal. Please review and forward to Brett Blacker. Ken Stowers was the

technical advisor. Check flats need more pushing to move up in queue. List of SI proposals is apparently too long to be met, so if you don't push, your proposals won't get done. John Campbell handles this prioritization.

Need letter to Roger Doxey stating that TEC's should remain on by default (see Stowers).

Gallagher is looking into the possibility that the TEC's could stay on during a software safing event -- important to preserve a UV flood.

I arranged two check-out badges for our team that will allow a person into the STOCC -- necessary for picking up tapes and attending meetings. Jon has one right now. See him about getting the other signed out (to Bob Light?).

I promised image info packets to Art Bradley and Domenic Tenerelli, who want to use image size, shape and drift for insight into HST pointing control. The relevant images so far are 5/31, 6/03, 6/04, 6/08. Desired info includes length and orientation of trail, if any, FWHM in two coordinates, mosaic of good images (enlarged), drift rate between two exposures. Jon is getting together material that will help, so consult with him. Confirm that they still need this stuff after you give them this first batch -- it's considerable work.

There are various reports of bias drift. Maybe just on early frames, when TEC's were warm. Should check on recent frames.

Check with Burrows to make sure that he changed the position of the primary target in the Burrows Westphal proposal to place the bright star at the center of one of the chips. He was supposed to fix this error.

Ed Groth is coming on Sunday for a few days. Will probably have to leave mid-week. Might be able to come back later.

Art Vaughn is on East Coast through Wed. Could come here Thurs. Communicate with him through Dave Rogers at JPL.

Tod is at Aspen. Stands ready to come and help if asked.

Kalinowski volunteered to provide scheduling information about TDRSS and SAA if we could not get help at the STScI.

All the recent faxes (Jewels and yours) are in my office on the windowsill, along with a complete set of graphic timelines and daily schedules. Ken Stowers has a complete set of SMS Timelines -- very useful!

5) Points for PMA Meeting on Tuesday: covers May 30 - June 12 for us due to missed meeting last week. Jon H. can help you with graphical materials. He can also help you understand the graphics in the two folders full of materials which are under this memo.

a) The WFPC has taken the following pictures, all in F547M:

May 30 PC-first light 10s, 100s; two faint stars only

	NGC 188 field	
May 31	WF real-time NGC 188 field	10s, 100s; on gyros; 11 good stars several faint galaxies; telescope oscillation of +/-0.2" present with period short compared to 10 s; images same (trailed) in both frames; kidneys visible moving around (to practiced eye), symmetric about center of each camera chip.
Jun 01	PC NCC 188	10s, 100s; blank field, no stars.
Jun 03	PC iota Car	10s, 100s; first coarse-lock guiding; one bright star upper left of PC5. Bright core contains 12% of total light (within 25 pixels). Asymmetric kidney visible to upper left, pointed away from center of chip. Mag of ~15.0 (using TV6 QE) seems about right.
Jun 04	PC iota Car	10s, 100s; telescope motion of 3.2" from previous frame moved bright star onto upper right corner of Chip 6 (sky coords.). Orientation of kidney flipped over, supporting hypothesis that kidneys are directed mainly away from centers of chips.
Jun 08	PC iota Car	10s, 100s; planned slew to place bright star in center of Chip 5 did not take place. Bright star still in upper left corner of Chip 5. Diplaced 0.39" from prev. position. Otherwise similar to first exposure.
Jun 10	PC iota Car	10s; 100s; ???????????? What happened?

b) Team activities during this period:

* Optical analysis of pictures to date. Information provided to Project and STScI as requested. Participation in daily 11:00 Project Status meeting.
* Astrometric position for OTA axis (V1) on WF 5/31 frame of NGC 188, which verified FHST position with error of 21" in RA and 9" in Dec.
* Informational support for optical modelling of aberrations provided to Lyon (PE) and Burrows (STScI). (What this means in reality is that I've shot down all their suggestions, except that Lyon hasn't accepted yet that his secondary misalignment theory fails to reproduce the observed kidney symmetry.)
* Positional and image parameters provided (on a preliminary basis) to Tenerelli and Bradley for information on spacecraft pointing and stability.
* SMS planning support.
* Construction of detailed exposure sequences for inclusion in SMS's.

c) Conclusions to be drawn from picture analysis (I have included a few personal comments for your benefit that are not intended for the report):

* Camera throughput in WF and PC seems nominal, but is not yet measured precisely.
* The images consistently have a core-halo morphology, with a third region of "tendrils" farther out.

  - 12% light in core.
  - 60% light in halo.
  - 30% light in outer tendrils (within 1" radius).

* At most field locations, the halo is incompletely filled, yielding a deformed, "kidney" shape. The lobe of the kidney is located approximately on the side of the image away from the center of the field of each chip. So far, we can't see any clear difference between using chip coords. and sky coords. If there was exact circular symmetry, there wouldn't be any difference. This is a good test for future data.

* Most likely hypothesis: change in shape of kidney is due to interaction or vignetting of a misaligned and/or aberrated OTA beam with WFPC obstructions and/or aperture stops.

  [Jim: I don't have any conclusive model yet myself. However, the situation I'm worrying about most is the following: large amounts of spherical aberration and some defocus. This reproduces the basic kidney morphology and circular symmetry of the kidneys around the center of the field. Quantitatively, it has a hard time accounting for the fact that the kidneys seem at be as prominent in the PC as in the WF, so Burrows doesn't like it. One can "cure" this by positing that the outer part of the primary is OK and makes the sharp core of the image (plus some of the light in the wings due to diffraction). The moving obscuration of the WFPC secondaries is then a larger fraction of the remaining light coming from the inner part of the entrance pupil only. You are then lead to a bizarre primary figure -- accurate in the outer parts but with severe spherical aberration (due to miscalculation of the shape change in going to zero-g?) in the inner regions. I haven't told this to anyone else but Jon and Roger, but it keeps me awake at night. That's why I'm interested in the PM actuators and what they can do by way of spherical aberration.

  Note that it seems to me that the outer part of the primary must be working, since otherwise how could we get the sharp core with width about equal to the diffraction limit of the telescope? Beware of ray traces that show sharp cores formed by the inner portions of the pupil -- these cores would disappear, I think, using physical optics.]

* A key point that must be matched by any model of the image: symmetry around the chip centers rather than constant orientation on the sky. (Private remark: I have stressed that I think this argues against a merely tilted OTA secondary, and therefore against Lyon's current model.)

* A key need: To obtain better populated star field with stars at the bottoms of the chips (in sky coords.).

* The tendrils change their appearance depending on position in the field. They merge smoothly with structures that are clearly

diffraction patterns in
the center of the image, lending support to the notion that they, too are diffraction phenomena. This could be tested with images at shorter wavelength. {Note: Burrows thinks these are diffraction caused by micro-roughness and/or medium-scale surface ripple. I favor the notion that they are caused by low-order aberrations such as coma and astigmatism.]

* There has been no discernible OTA focus change since First-Light. the amount of light in the core of the image has remained constant at ~12-15%. Defocus due to degassing is small.